5V/5A
Fsw=600kHz
L=2u2H
Ic(RMS1) = 2A

D = Vout/Vin = 5/12.6 = 0.397
D = Vout/Vin = 5/9.6 = 0.521
Using 9.6V since its closer to 50% duty

I_Ripple = (Vout(1 - Vout/Vin))/(L*f) ; use larger Vin
I_In = Iout * (Vout/Vin)
Ic(RMS) = Iout * SQRT[(Vout/Vin)(1-(Vout/Vin))]
Ic(RMS)tot = SQRT[Ic(RMS1)^2 + Ic(RMS2)^2]

I_Ripple = 2.285A
I_In = 2.6A
Ic(RMS2) = 2.497A
Ic(RMS)tot = 3.2A -> 3.8A

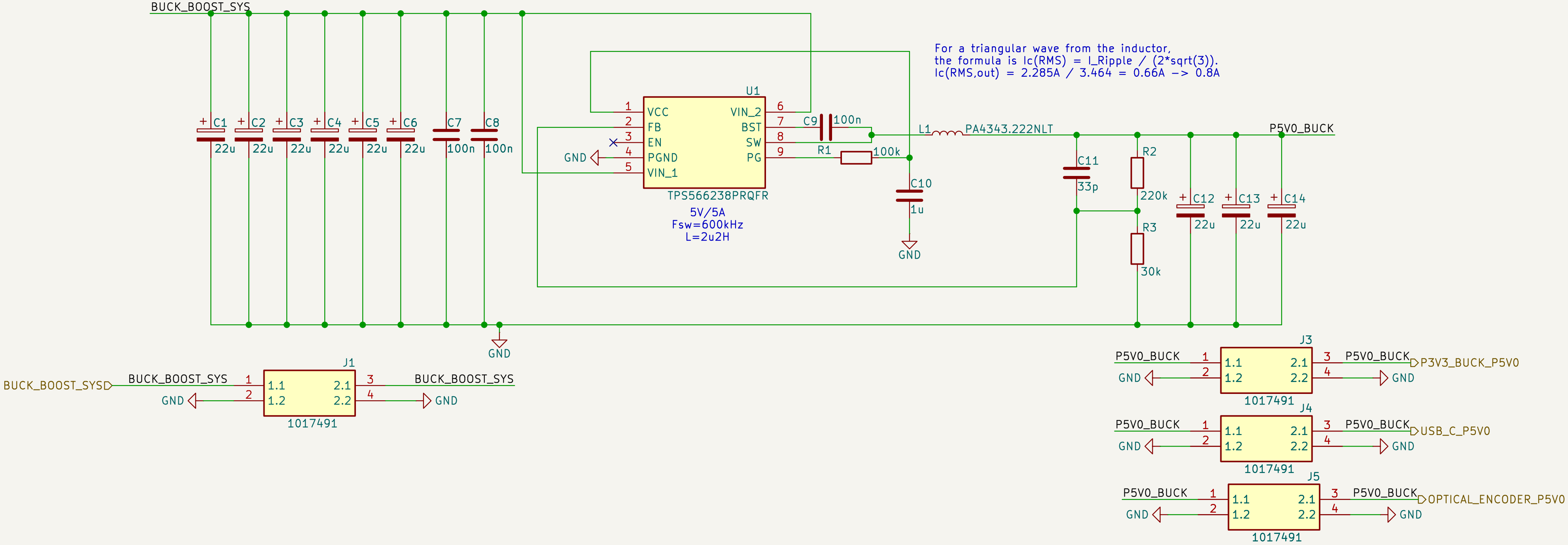
1, find cap bank > 3.8A rms ripple current

2, Ripple_Vout = I_Ripple * ESR
50mA = 2.285 * ESR, ESR=21.88mOhm

3, 2.6A is drawn from the previous stage

5V Buck

For a triangular wave from the inductor,
the formula is Ic(RMS) = I_Ripple / (2*sqrt(3)).
Ic(RMS,out) = 2.285A / 3.464 = 0.66A -> 0.8A



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